

MULTIMODAL EMOTION RECOGNITION USING SPEECH AND TEXT

1. Introduction

Emotion recognition is an important task in human-computer interaction and artificial intelligence. This project aims to identify emotions using speech signals, text data, and multimodal fusion techniques.

The system predicts seven emotions:

- Angry
- Disgust
- Fear
- Happy
- Neutral
- Surprise
- Sad

Dataset Used:

Toronto Emotional Speech Set (TESS)

2. Preprocessing

Speech Preprocessing:

- Audio files were loaded using Librosa.
- Sampling rate was standardized to 16 kHz.
- MFCC (Mel Frequency Cepstral Coefficients) features were extracted.
- Variable-length audio signals were padded/truncated to fixed dimensions.

Text Preprocessing:

- Emotion labels were cleaned and standardized.
- Text inputs were tokenized using DistilBERT tokenizer.
- Input IDs and attention masks were generated for transformer processing.

3. Architecture Decisions

Speech Pipeline:

Audio

- MFCC Feature Extraction
- BiLSTM
- Dense Layer
- Emotion Prediction

Reason:

MFCC captures important speech characteristics such as pitch and frequency, while BiLSTM captures temporal dependencies in speech sequences.

Text Pipeline:

Text

- DistilBERT
- Dense Layer
- Emotion Prediction

Reason:

DistilBERT efficiently captures contextual and semantic information from textual input.

Fusion Pipeline:

Speech Features

+

Text Features

- Concatenation
- Dense Layer
- Emotion Prediction

Reason:

Fusion combines complementary information from multiple modalities for improved emotion recognition.

4. Experimental Results

Speech-only Accuracy:

99.82%

Text-only Accuracy:

12.5%

Fusion Accuracy:

99.82%

5. Analysis

The speech-only model achieved very high accuracy because emotional information is strongly present in speech signals through pitch, tone, energy, and frequency variations.

The text-only model showed weak performance because the TESS dataset contains repeated neutral words with limited emotional meaning.

Initially, the fusion model showed weak performance. Later improvements included:

- using pretrained speech encoder weights
- fixing train/test split consistency
- reducing train-test overlap risks
- improving fusion training configuration

These improvements significantly increased fusion performance.

The pretrained speech encoder played a major role in improving the fusion model because emotional information in TESS is primarily carried through acoustic signals rather than textual semantics.

Among the emotional classes, Neutral, Sad, and Disgust were easiest to classify because their speech patterns were highly distinguishable in the dataset.

Fear and Angry occasionally showed minor confusion because both emotions can share similar acoustic intensity and energy patterns.

The final fusion model achieved near-perfect accuracy on the TESS dataset.

6. Confusion Matrix Analysis

Most emotional classes were classified correctly with near-perfect precision and recall.

Only a very small number of samples were misclassified.

The confusion matrix demonstrated strong separability between emotional classes.

7. t-SNE Analysis

t-SNE visualization was used to analyze the learned feature representations produced by the models.

Distinct emotion clusters were observed in the learned feature space.

The temporal modelling block (BiLSTM), contextual modelling block (DistilBERT), and fusion representations showed good separability among emotional classes.

This indicates that the models learned discriminative emotional representations effectively.

8. Future Improvements

- Real-time emotion recognition
- Attention-based fusion methods
- Transformer-based speech models
- Better multimodal conversational datasets
- Real-time web deployment